

Minimum Wage Productivity and Inflation in Portugal

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Abstract

This paper studies the relationship between statutory minimum-wage policy, labour productivity, and consumer-price inflation in Portugal from the introduction of the national minimum wage in 1974. The long-run historical analysis is combined with an exposure-based regional and sectoral design intended to identify the dynamic pass-through of minimum-wage-induced labour-cost shocks into consumer prices. The empirical design separates statutory wage changes from productivity growth and prior inflation, exploits heterogeneity in minimum-wage coverage, and controls for taxes, energy, imported inflation and macroeconomic regimes.

1 Introduction

2 Institutions

3 Literature

4 Data

All series are retrieved programmatically, checksummed, and rebuilt from raw inputs. No value in this paper is typed by hand: every number in the text is a macro generated from the analysis dataset, and every table and figure is written by the pipeline that produced it.

4.1 Statutory minimum wage

The statutory series is parsed from the official history published by the Direção-Geral do Emprego e das Relações de Trabalho (DGERT), which lists one row per legal act with its effective date and the value it set. The series begins with Decreto-Lei n.º 217/74 of 27 May 1974, which introduced a minimum wage of 3 300 escudos, and runs to the present.

Portugal did not have a single minimum wage until 2005. Up to 1991 three legally distinct wages coexisted — domestic service, agriculture, and a general regime covering the remaining activities — falling to two between 1992 and 2004. The general regime is the series conventionally quoted as the Portuguese minimum wage, and it is the one used throughout. The special regimes were always set at or below it, which the pipeline asserts on every date where regimes coexist.

Escudo amounts are converted at 200.482 escudos per euro, the irrevocable rate fixed for the changeover. The act effective in January 2002 states both currencies, and its published euro figure is preferred over the converted one.

Two independent checks validate the parsed series. First, the provider states each act's percentage increase over the wage it replaced; recomputing that from consecutive parsed values reconciles amounts, ordering, and regime assignment simultaneously. Second, Eurostat compiles the same statutory wage from its own reading of national law. Eurostat expresses it on a

twelve-month basis whereas Portuguese law states a monthly wage paid fourteen times a year, so its figure equals the monthly level times 14/12; undoing that convention reproduces the parsed series to within a euro across the years checked.

The reconciliation exposes one gap in the national source: the act effective in 2000 is absent from the published history, while the 2001 entries still state their increase relative to it. Because a year with no listed act is otherwise indistinguishable from a year in which the wage was genuinely frozen — as it was in 2012 and 2013 — the pipeline resolves the ambiguity against Eurostat, and uses the Eurostat level only where the two compilers materially disagree. The 2000 observation is therefore attributed to Eurostat rather than to the national legal source, and is flagged as such in the dataset.

4.2 Prices and productivity

Consumer prices are the World Bank index for Portugal, available from 1960. Eurostat’s harmonised index begins in 1996, two decades after the minimum wage was introduced, and would truncate the historical layer at the point where it is least informative.

Labour productivity is real gross domestic product per person employed from the European Commission AMECO database (series `PRT.1.1.0.0.RVGDE`), available from 1960. The Eurostat national-accounts equivalent begins in 1995 for Portugal. Over the 31 years in which the two overlap they are effectively the same series: their annual growth rates correlate at unity and differ by at most 0.003 percentage points. AMECO carries Commission projections beyond the last outturn, which are excluded; only observed years enter the dataset.

4.3 Coverage and exposure

The exposure design requires the pre-existing share of workers paid at the minimum wage, by region and industry. The Gabinete de Estratégia e Planeamento (GEP) published these statistics, but has since withdrawn the host: every path under its former domain now answers with an unrelated landing page. The publications are retrieved instead from the Internet Archive and are registered with their archival addresses and checksums, so the citation remains verifiable.

The recovered monitoring reports give coverage by economic activity. They do not give a systematic breakdown by NUTS II region, which the identification strategy in Section ?? requires. Constructing a region-by-industry measure from these sources would require assuming that within-industry coverage does not vary across regions, so that all regional variation comes from industry mix. That assumption is not innocuous in a country whose regions differ sharply in tourism intensity, and it is not made here. The price pass-through estimates are therefore not reported in this version.

5 Methods

6 Long-run evidence

This section reports the historical accounting layer. It is descriptive. It documents what the statutory wage did relative to a productivity-and-inflation benchmark, and motivates the identification strategy; it is not itself evidence of a causal effect of minimum wages on prices.

6.1 The benchmark and the residual

For year t the benchmark nominal wage growth compounds productivity growth with the previous year's inflation,

$$g_t^{\text{benchmark}} = (1 + g_t^A)(1 + \pi_{t-1}) - 1, \quad r_t = g_t^{MW} - g_t^{\text{benchmark}}, \quad (1)$$

where r_t is the policy residual. The benchmark compounds rather than adds. The distinction is immaterial at recent inflation rates and is not immaterial over the sample: with inflation near thirty per cent, as in the early 1980s, the additive approximation understates the benchmark by more than a percentage point, which exceeds most of the residuals being measured.

6.2 The wage floor against productivity

Figure 1 indexes the real minimum wage and output per worker to 1974. Over the 52 years to 2025, productivity reached 257.0 while the real minimum wage reached 128.3.

The divergence is the central historical fact. The real value of the wage floor fell sharply through the inflation of the late 1970s and early 1980s and did not regain its 1974 level for four decades. Measured against productivity, the ratio bottomed at 34.6 in 2007 and stood at 49.9 at the end of the sample: the minimum wage buys a far smaller share of what an average worker produces than it did when it was introduced.

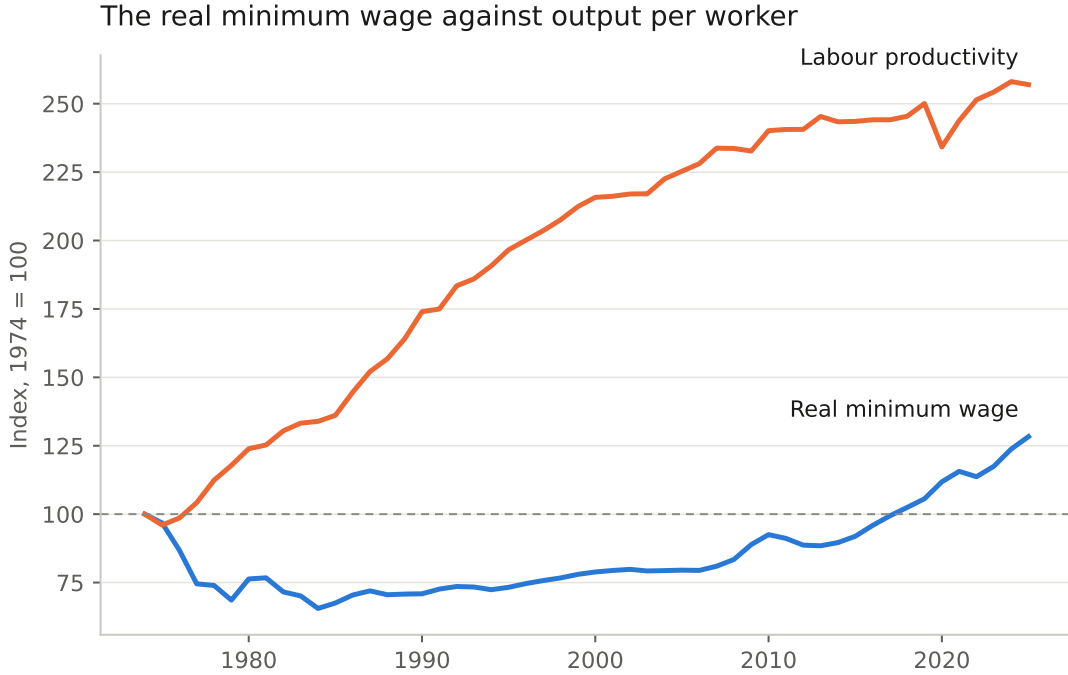


Figure 1: Real minimum wage and labour productivity, indexed to 1974. Both series are on one axis; productivity is real GDP per person employed.

6.3 The residual by regime

Averaged over the sample the residual is -2.05 percentage points a year, and cumulates to -102.4 points. The average conceals a reversal, and Table 1 shows it.

Three periods stand out. Through the revolutionary transition and its aftermath the statutory wage rose quickly in nominal terms and yet fell far behind a benchmark driven by inflation above twenty per cent a year; the residual is strongly negative and the real wage floor



Figure 2: The real minimum wage per unit of labour productivity, indexed to 1974.

Table 1: Annual averages by policy regime, in per cent. π is consumer price inflation, g_A productivity growth, g_{MW} statutory minimum-wage growth, and the residual is defined in equation (1).

Regime	Years	π	g_A	g_{MW}	Residual
revolutionary_transition	1974–1985	21.7	2.9	17.5	-8.2
european_integration	1986–1998	8.0	3.3	9.1	-3.9
euro_pre_crisis	1999–2009	2.6	1.0	4.0	-0.0
sovereign_debt_crisis	2010–2014	1.6	0.9	1.7	-0.6
minimum_wage_restart	2015–2019	0.8	0.5	4.1	+2.9
pandemic_and_inflation	2020–2026	3.0	0.5	6.4	+3.2

lost roughly a third of its value. From the late 1990s the residual is close to zero: the wage floor broadly tracked productivity and prices. From 2015 the sign reverses, and the residual turns persistently positive, which is the policy shift the identification strategy is designed to exploit.

Figure 3 plots the annual series.

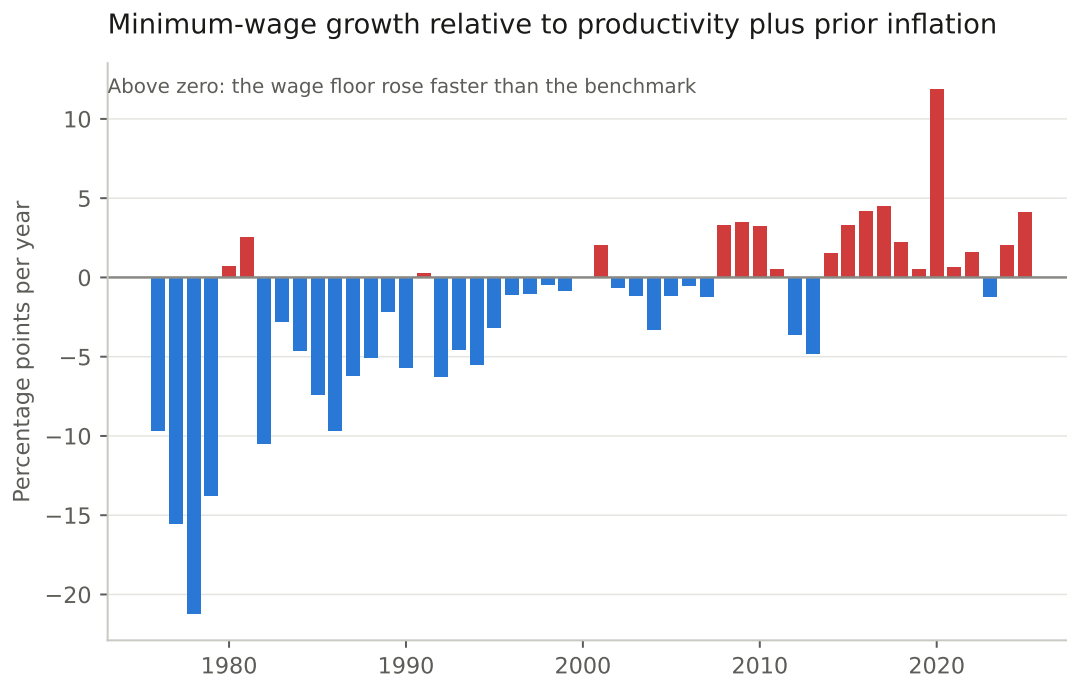


Figure 3: Annual policy residual: statutory minimum-wage growth less productivity growth compounded with prior inflation.

6.4 What this does not establish

The residual is an accounting object. A positive residual means the wage floor rose faster than the benchmark; it does not mean the increase was inflationary, and the correlation between the residual and subsequent inflation cannot settle the question, because statutory changes respond to the same conditions that drive prices. That is precisely why the causal layer uses predetermined exposure and panel variation rather than the national time series, and why no pass-through estimate is reported until the regional coverage data required by that design is available.

7 Robustness

8 Conclusion

References